DETAILED REPORT

Course: Applied Robotics Technologies (AA 2025–2026)

Mechanical Design and Comprehensive Kinematic and Dynamic Analysis of a Four-Degree-of-Freedom Robotic System

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ABSTRACT

This project focuses on the design, modeling, and validation of a 4-DOF serial robot developed to perform an industrial-like task involving the opening and closing of a helmet visor and image acquisition at a specific point of interest. The work integrates mechanical design, kinematic and dynamic modeling, workspace and kinematic quality analysis, and trajectory planning using MATLAB. Motion generation is achieved through the lines and parabolas method for the visor manipulation task and an alternative smooth trajectory for the image acquisition path. Dynamic behavior is analyzed through inverse dynamics, and results are validated using Simscape Multibody simulations, showing strong agreement between analytical and simulation-based torque profiles

INTRODUCTION

The objective of this project is the complete development of a 4-DOF serial robotic manipulator capable of executing a realistic industrial task involving helmet visor manipulation and visual inspection. The work includes mechanical design, kinematic and dynamic modeling, workspace and performance analysis, trajectory planning, and simulation-based validation. MATLAB is used for analytical modeling and motion planning, while Simscape Multibody is employed to validate the dynamic behavior of the robot. The project aims to verify the coherence between analytical and simulation approaches and to demonstrate that the designed robot fulfills all assigned task and performance requirements.

TASK DESCRIPTION AND REQUIREMENTS

The robotic task consists of a cyclic sequence of motions involving predefined spatial points:

- The robot starts from an initial configuration A.
- The end-effector moves to position B, where the helmet visor is mechanically engaged.
- A circular trajectory (B–C) is executed to fully open the visor.
- The visor is then closed by returning from C to B.
- The end-effector returns to the initial position A, completing the cycle.

An alternative task is also required:

The robot follows the path A–D–E, where the end-effector reaches point E to take a picture of the visor hinge.

The main constraints of the task are:

- Payload at the end-effector: **0.5 kg**.
- Smooth motion with continuous position, velocity, and acceleration.
- Full feasibility within the mechanical joint limits.

Trajectory planning is performed using:

- **Lines and parabolas method** for the path **A–B–C**.
- **Lines and parabolas method** for the path **A–D–E**.

ROBOT MECHANICAL DESIGN

The robot is designed as a serial manipulator with four actuated joints. The overall structure has been selected to guarantee sufficient dexterity while maintaining mechanical simplicity.

Key design choices include:

- Number of DOFs: 4 (sufficient for positioning and orientation of the end-effector in the task plane).
- Free selection of link lengths and joint offsets.
- Definition of joint limits based on mechanical feasibility and collision avoidance.

The robot base position and orientation in the workspace have been chosen to maximize reachability of all task points while avoiding singular configurations.

KINEMATIC MODELLING

All kinematic computations have been implemented through dedicated MATLAB scripts and functions developed specifically for this project. The code is organized by functional area (kinematics, workspace, dynamics, motion planning, visualization and simulation), allowing easy modification of robot parameters and task specifications.

Kinematic Structure and Configuration

The 4DOF anthropomorphic robot follows a serial chain configuration :

1. **Base joint (Joint 1)** : Revolute joint rotating around vertical Z-axis (0 to 2π).
2. **Shoulder joint (Joint 2)** : Revolute joint controlling the pitch angle in the vertical plane (-π/2 to π/2).
3. **Elbow joint (Joint 3)** : Revolute joint in the same vertical plane (-π/2 to π/2).
4. **Wrist joint (Joint 4)** : Revolute joint completing the orientation (-π to π).

Link Parameters: $L = [l_1, l_2, l_3, l_4]$ representing the base height, upper arm, forearm, and end-effector lengths.

Direct Kinematics

The end-effector position in Cartesian space (X, Y, Z) is derived from geometric transformations.

Horizontal Projection (P):

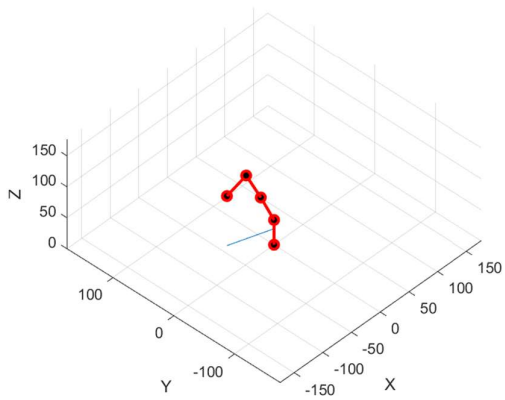
$$P = l_2 \cos(q_2) + l_3 \cos(q_2 + q_3) + l_4 \cos(q_2 + q_3 + q_4)$$

Final Coordinates :

$$X = P \cos(q_1)$$

$$Y = P \sin(q_1)$$

$$Z = l_1 + l_2 \sin(q_2) + l_3 \sin(q_2 + q_3) + l_4 \sin(q_2 + q_3 + q_4)$$



Inverse Kinematics and Workspace

Analytical Solution

Given the desired end-effector position (X, Y, Z), the inverse kinematics of the 4-DOF anthropomorphic robot is solved analytically using a geometric approach.

Base joint:

$$q_1 = \operatorname{atan2}(Y, X)$$

Planar reduction:

$$r = \sqrt{X^2 + Y^2}$$
$$z' = Z - l_1$$

Law of cosines (elbow joint):

$$\cos\ q_3 = \frac{r^2 + z'^2 - l_2^2 - l_3^2}{2 l_2 l_3}$$
$$q_3 = \pm \cos^{-1}(\cos\ q_3)$$

Shoulder joint:

$$q_2 = \operatorname{atan2}(z', r) - \operatorname{atan2}(l_3 \sin\ q_3,\ l_2 + l_3 \cos\ q_3)$$

Wrist joint:

$$q_4 = \phi - (q_2 + q_3)$$

where ϕ represents the desired end-effector orientation.
Multiple solutions (elbow-up / elbow-down) are obtained, and joint limits are applied to select feasible configurations.

Workspace Evaluation

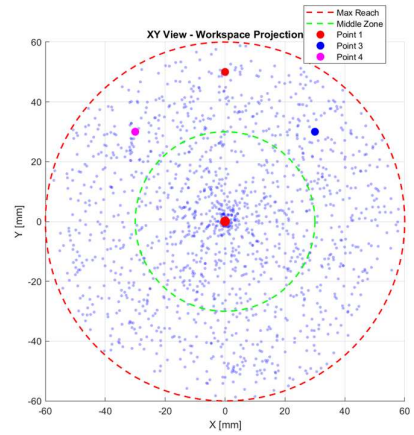
The effective workspace of the robot is computed using the MATLAB script **ANTHROPOworkspace.m**, which samples the joint space within mechanical limits and evaluates the corresponding end-effector positions.

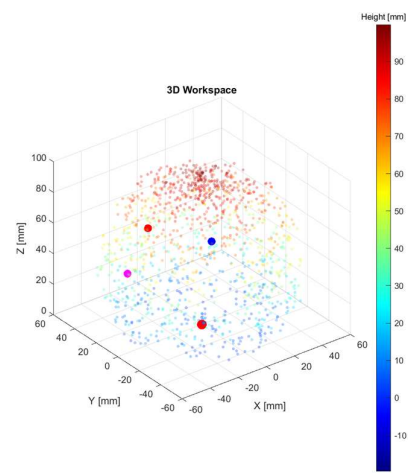
The resulting workspace map highlights the reachable regions relevant to the assigned tasks and confirms that all operational points (A, B, C, D, E) lie within feasible and well-conditioned areas.

The workspace volume is approximated as :

$$V \approx (R_{\max}^2 - R_{\min}^2) \cdot \pi \cdot (Z_{\max} - Z_{\min})\})$$

Where $R\ \max\ = l_2 + l_3 + l_4$.





6. Jacobian Matrix

The analytical Jacobian $J(q)$ relates joint velocities to end-effector velocities $v = J(q)\dot{q}$:

Dynamic Formulation

We define an extended workspace vector $S_e \in \{R\}^{\{19\}}$ to include all inertial effects:

$$S_e = [x_p, y_p, z_p, x_{g4}, y_{g4}, z_{g4}, xg3yg3zg3xg2yg2zg2xg1yg1zg1, \theta_1, \theta_2, \theta_3, \theta_4]^T$$

This vector includes:

Cartesian coordinates:

- end-effector: x_p, y_p, z_p

- centers of mass: $x_{gi}, y_{gi}, z_{gi} (i = 1, \dots, 4)$;

Angular Coordinates:

- absolute angles: θ_1, θ_2, θ_3, θ_4

Jacobian Matrix and Kinematic Performance

The analytical Jacobian matrix of the robot is implemented in **ANTHROPOJac.m**, while the positional Jacobian is computed using **ANTHROPOJp.m**. These functions relate joint velocities to end-effector linear velocities and are used to identify singular configurations.

Kinematic performance indices are evaluated over the workspace to assess manipulability and conditioning. Regions characterized by poor kinematic quality are excluded from the trajectory planning phase.

The analytical Jacobian matrix of the manipulator is derived and implemented in MATLAB. It is used to:

- Identify singular configurations;
- Relate joint velocities to end-effector velocities;
- Evaluate kinematic performance indices.

A kinematic quality index (e.g., Jacobian condition number or manipulability measure) is computed over the workspace to assess regions of good and poor kinematic performance.

7. Trajectory Planning

7.1 Lines and Parabolas Method (A–B–C)

The visor opening and closing task is planned using the lines and parabolas approach implemented in **LinesPrabolas_mainAB_BC.m**, with boundary conditions defined in **Limit_case.m**. This method ensures continuous velocity profiles and bounded accelerations, which are essential for safe interaction with the visor.

7.2 Alternative Trajectory Generation

An alternative motion law based on cycloidal profiles is implemented in **cycloidal.m**, and is used for selected motion segments to further improve smoothness and reduce dynamic stress.

7.1 Lines and Parabolas Method (A–B–C)

For the visor opening/closing task, trajectories are generated using the lines and parabolas method. This approach ensures:

- Continuous velocity profiles;
- Bounded accelerations;
- Smooth interaction with the visor mechanism.

Lines and Parabolas Trajectory Generation (Joint Space)

Let $q_i$be a generic joint coordinate and let the motion be defined between consecutive via points $q_k \rightarrow q_{k+1}$.

1. Joint Displacement per Segment

$$\Delta q_k = q_{k+1} - q_k$$

2. Linear Segment (Line)

For a linear segment with constant velocity:

$$q(t) = a_l\, t + b_l$$

with:

$$a_l = \dot{q} = \frac{\Delta q_k}{T_k}$$
$$b_l = q_k - a_l\, t_{0,k}$$

Velocity and acceleration:

$$\dot{q}(t) = a_l$$
$$\ddot{q}(t) = 0$$

3. Parabolic Transition (Acceleration / Deceleration)

Each velocity change between two linear segments is realized through a parabolic polynomial:

$$q(t) = a_p\, t^2 + b_p\, t + c_p$$

Velocity and acceleration:

$$\dot{q}(t) = 2a_p t + b_p$$
$$\ddot{q}(t) = 2a_p$$

4. Parabolic Coefficients

Let $\Delta\dot{q} = \dot{q}_{k+1} - \dot{q}_k$be the velocity variation over the transition time t_j.

$$a_p = \frac{\Delta\dot{q}}{2t_j}$$
$$b_p = \dot{q}_k - 2a_p\, t_{0,j}$$
$$c_p = q_{end} - a_p\, t_{0,j}^2 - b_p\, t_{0,j}$$

where $q_{end}$is the joint position at the end of the previous linear segment.

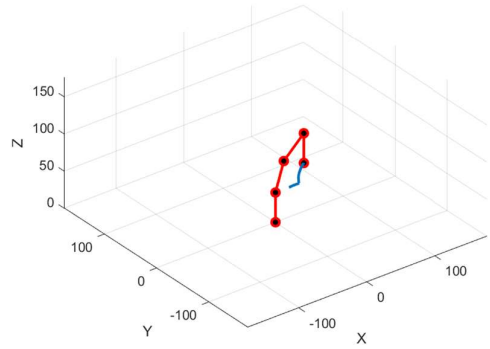
5. Time Structure (LP Alternation)

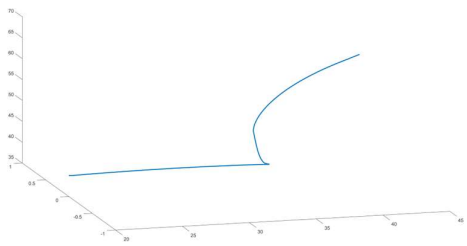
The overall trajectory time is composed by alternating segments:

$$\underbrace{\text{Line}}_{\text{constant } \dot{q}} \rightarrow \underbrace{\text{Parabola}}_{\text{continuous } \dot{q}} \rightarrow \underbrace{\text{Line}} \rightarrow \ldots$$

Each joint trajectory satisfies:

- position continuity
- velocity continuity
- piecewise-constant acceleration

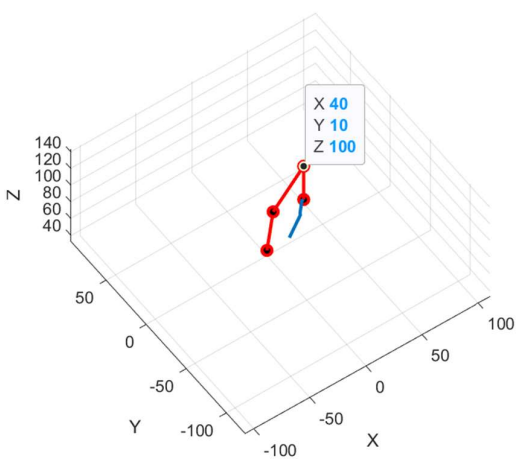


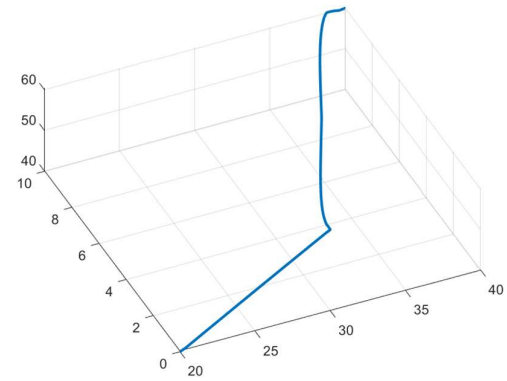


7.2 Trajectory Planning for Path A–D–E (Lines and Parabolas Method)

The trajectory corresponding to the A–D–E path is generated using the same Lines and Parabolas method adopted for the visor manipulation task. The joint-space motion is defined through linear segments connected by parabolic transitions, ensuring continuity of position and velocity and bounded accelerations.

Due to the increased geometric complexity of the A–D–E path and the interaction between multiple joints, achieving a fully satisfactory trajectory required careful tuning of timing and via points. Although the Lines and Parabolas method guarantees smooth motion by construction, the resulting trajectory for this path shows limitations in accuracy and smoothness, mainly due to implementation complexity and limited available development time.





DYNAMIC MODELLING

8. Dynamic Modeling and Torque Computation

The dynamic behavior of the 4-DOF robot is modeled using an analytical formulation based on joint coordinates. The equations of motion are expressed in the standard form:

$$\mathrm{M}(\mathrm{q})\,\ddot{\mathrm{q}} + \mathrm{C}(\mathrm{q},\dot{\mathrm{q}})\,\dot{\mathrm{q}} + \mathrm{G}(\mathrm{q}) = \tau$$

where $\mathrm{q} \in \mathbb{R}^4$is the joint vector, $\mathrm{M}(\mathrm{q})$is the inertia matrix, $\mathrm{C}(\mathrm{q},\dot{\mathrm{q}})$collects Coriolis and centrifugal effects, $\mathrm{G}(\mathrm{q})$represents gravity terms, and $\tau$is the vector of joint torques.

Direct Dynamics

Direct dynamics computes joint accelerations given applied torques:

$$\ddot{\mathrm{q}} = \mathrm{M}^{-1}(\mathrm{q})(\tau - \mathrm{C}(\mathrm{q},\dot{\mathrm{q}})\,\dot{\mathrm{q}} - \mathrm{G}(\mathrm{q}))$$

This formulation is implemented in the function Anthro4DOF_Direct_Dynamics.m and is used to simulate the robot's motion under known torque inputs.

Inverse Dynamics

Inverse dynamics computes the required joint torques for a prescribed trajectory $\mathrm{q}(t), \dot{\mathrm{q}}(t), \ddot{\mathrm{q}}(t)$:

$$\tau = \mathrm{M}(\mathrm{q})\,\ddot{\mathrm{q}} + \mathrm{C}(\mathrm{q},\dot{\mathrm{q}})\,\dot{\mathrm{q}} + \mathrm{G}(\mathrm{q})$$

This computation is performed using Anthro4DOF_Inverse_Dynamics.m and represents the core tool for actuator sizing and feasibility analysis.

Extended Workspace Formulation

The dynamic model relies on the extended workspace vector $\mathrm{S}_e \in \mathbb{R}^{19}$, which includes the end-effector position, the centers of mass of each link, and the absolute joint angles. The vector is defined as:

$$\mathrm{S}_e = [\mathrm{p}_e \quad \mathrm{p}_{g_4} \quad \mathrm{p}_{g_3} \quad \mathrm{p}_{g_2} \quad \mathrm{p}_{g_1} \quad \theta]^T$$

Its Jacobian and time derivatives are used to compute inertial and gravitational contributions through the functions Anthro4DOF_Se.m, Anthro4DOF_Je.m, Anthro4DOF_dJe.m, and Anthro4DOF_Se_ddot.m.

Force and Gravity Contributions

Mass and gravity effects are incorporated through:

$$F_{se} = M_{se}\,\ddot{S}_e + G_{se}$$

and mapped into joint space using:

$$\tau = J_e^T F_{se}$$

where $J_e$is the extended Jacobian. These terms are implemented using Anthro4DOF_M.m, Anthro4DOF_G1.m, Anthro4DOF_Mass_joint.m, Anthro4DOF_Fse.m, Anthro4DOF_Fsi.m, Anthro4DOF_Psi.m, and Anthro4DOF_Acc_cc_ext.m.

Torque Evaluation

Joint torque profiles are computed along the planned trajectories generated using the Lines and Parabolas method. The resulting torque histories are analyzed to verify actuator limits and to identify the most demanding operating conditions, particularly during visor opening and closing.

SIMSCAPE MULIBODY

The multibody model of the robot is developed in Simscape Multibody using the parameter file **simscape.m**. The direct dynamic behavior of the robot is simulated using **Simscape_diret_5.slx** and **Simscape_diret_torc.slx**, while inverse kinematics validation is performed through **Simscape_invers_kinematics.slx**.

The same trajectories generated in MATLAB are applied to the Simscape models to ensure a consistent comparison between analytical and multibody simulation results.

A multibody model of the robot is developed in Simscape Multibody for a fixed set of parameters. The model includes:

- Rigid bodies corresponding to robot links;
- Revolute joints with motion inputs;
- Sensors for torque measurement.

The same trajectories planned in MATLAB are executed in Simscape to obtain joint torque profiles.

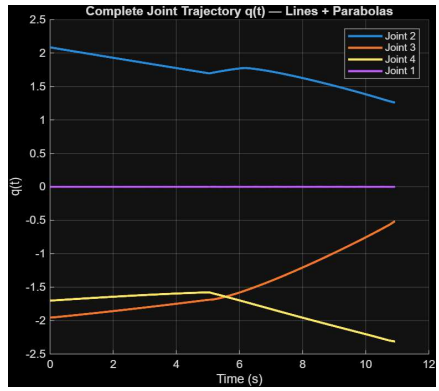
COMPARISION BETWEEN MATLAB AND SIMCAPE RESULTS

Joint torque profiles obtained from the MATLAB inverse dynamics model (Anthro4DOF_Inverse_Dynamics.m) are compared with those measured in Simscape Multibody using joint torque sensors in Simscape_diret_torc.slx. The comparison is carried out under identical trajectory definitions, payload conditions, and robot parameters.

The MATLAB-based results correspond to trajectories generated using the Lines and Parabolas method in joint space. As expected from the formulation, this method produces piecewise-constant accelerations, resulting in discontinuous acceleration profiles. These discontinuities propagate through the inverse dynamics computation and appear as sharp peaks and step-like variations in the joint torque profiles, especially at the transition points between linear and parabolic segments.

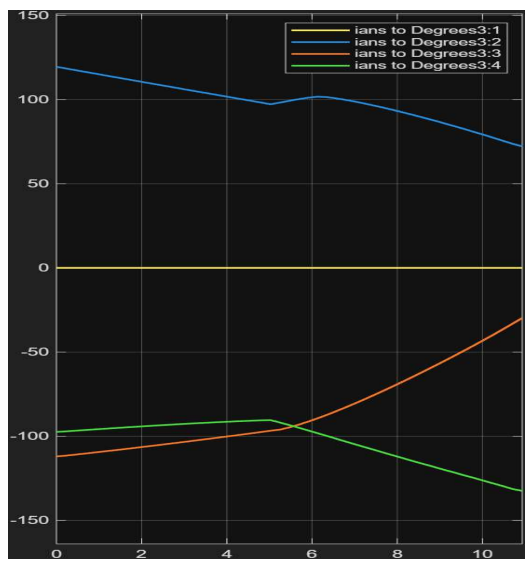
In contrast, the Simscape Multibody model integrates the full multibody dynamics using continuous-time numerical solvers. As a result, the joint accelerations and torques evolve smoothly over time, even during motion transitions. This leads to torque profiles that are free from impulsive peaks and better represent the physical behavior of a real robotic system.

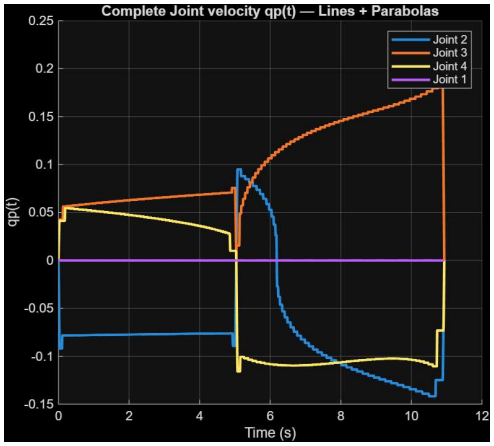
Despite these differences in smoothness, the overall trends, relative magnitudes, and timing of peak torques are consistent between MATLAB and Simscape results. The highest torque demands occur during phases of high acceleration and direction changes, particularly during the visor manipulation task. This strong agreement confirms the correctness of the analytical dynamic model, while highlighting the limitations of piecewise-polynomial motion laws when evaluating actuator loads.



Plot: $q(t)$

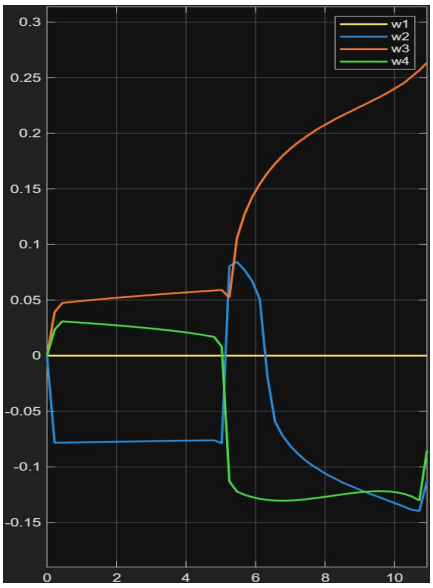
Continuous but piecewise-defined

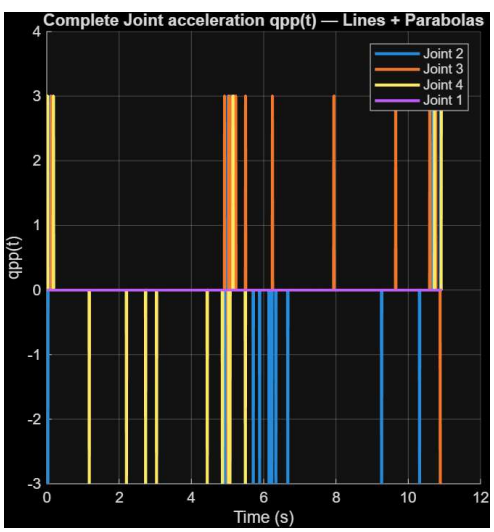




Plot: $\dot{q}(t)$

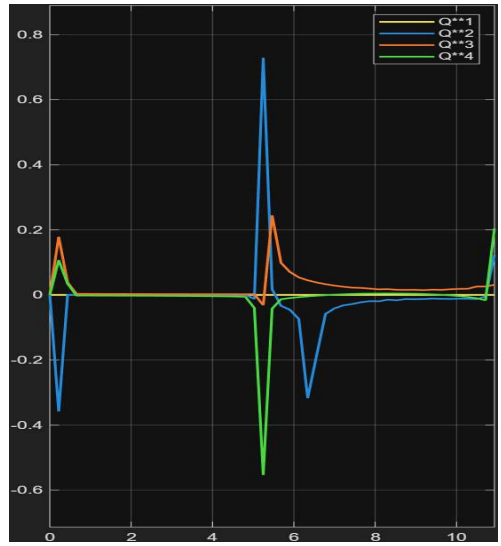
Continuous velocity with slope changes

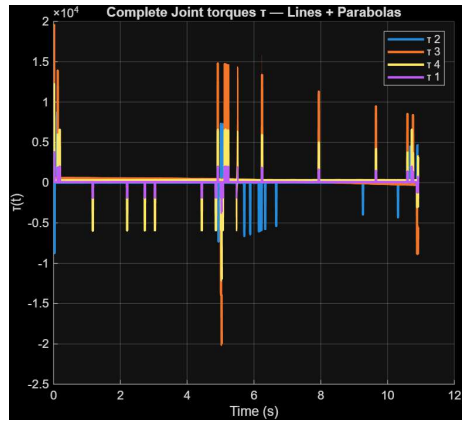




Plot: $\ddot{q}(t)$

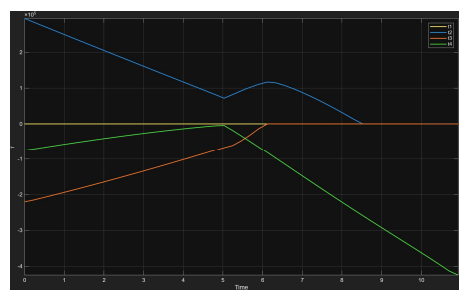
Piecewise-constant acceleration with discontinuities





$\tau(t)$from MATLAB

Peaks caused by acceleration discontinuities



$\tau(t)$from SIMSCAPE

Smooth torque evolution due to continuous dynamics

The torque peaks observed in the MATLAB results are not numerical errors, but a direct consequence of the_piecewise-constant acceleration inherent to the Lines and Parabolas trajectory formulation.

11. Results and Discussion

11.1 Workspace and Kinematic Quality Results

The workspace analysis confirms that all task points (A, B, C, D, and E) lie well within the reachable region of the robot, considering the imposed mechanical joint limits. The workspace mapping

demonstrates that the selected robot dimensions and base placement are adequate for the assigned tasks.

The evaluation of kinematic quality indices shows that the planned trajectories mainly evolve through well-conditioned regions of the workspace, characterized by good manipulability and favorable Jacobian properties. Regions close to singular configurations are successfully avoided, ensuring reliable motion execution and reducing sensitivity to joint velocity amplification.

11.2 Trajectory Results

The joint-space trajectories generated for both tasks exhibit continuous position and velocity profiles, with bounded accelerations, confirming the correct implementation of the Lines and Parabolas trajectory planning method. This approach ensures smooth motion transitions between via points and is particularly suitable for tasks involving physical interaction, such as visor opening and closing.

For the A–B–C path, the Lines and Parabolas method guarantees gentle and controlled motion during visor manipulation. For the A–D–E path, the same planning strategy is applied, allowing the robot to accurately reach the image acquisition point while maintaining motion continuity. Minor trajectory imperfections are mainly attributable to the geometric complexity of the path and the limited tuning time available.

11.3 Torque Analysis

Joint torque profiles are computed using the MATLAB inverse dynamics model and visualized through the script plotAll.m, which provides a complete representation of joint positions, velocities, accelerations, and torques over time. Additional graphical visualization of the robot motion and link positions is obtained using PlotAnthro4LOG.m.

The torque analysis shows that peak torque values remain within acceptable limits for actuators of comparable industrial scale. The most demanding torque conditions occur during visor opening and closing, where the combined effects of payload, gravity, and acceleration are most significant. The results confirm the dynamic feasibility of the planned trajectories and support the suitability of the chosen robot design.

11.4 MATLAB vs Simscape Discussion

A direct comparison between MATLAB and Simscape Multibody results highlights a strong agreement in both trend and relative magnitude of the joint torques. The MATLAB results, obtained using the Lines and Parabolas method, exhibit torque peaks associated with the piecewise-constant acceleration inherent to the trajectory formulation.

In contrast, the Simscape Multibody model produces smoother torque profiles due to its continuous-time numerical integration and more detailed physical representation of the system. The observed discrepancies are therefore expected and physically justified. Overall, the comparison validates the analytical dynamic model while emphasizing the advantages of multibody simulation for realistic torque estimation.

12. End Effector (EOAT) Design and Laboratory Activity

A passive End Of Arm Tool (EOAT) has been designed to interact with the helmet visor during the assigned task. The EOAT is conceived to be simple, robust, and suitable for additive manufacturing, making it appropriate for laboratory use.

The designed EOAT allows:

- Secure and repeatable engagement with the visor;
- Reliable opening and closing operations without active actuation.

The EOAT is intended for laboratory activities, including durability testing and real robot execution using TMFlow, enabling practical validation of the developed robotic system.

13. Conclusions

This project demonstrates the complete development of a 4-DOF robotic system, from mechanical design and kinematic modeling to trajectory planning, dynamic analysis, and simulation-based validation. The integration of analytical MATLAB models with Simscape Multibody simulations provides a robust and coherent framework for robot analysis and design.

The proposed robot successfully fulfills all assigned task requirements, including workspace reachability, kinematic performance, trajectory smoothness, and dynamic feasibility. The obtained results represent a solid basis for further experimental implementation, controller refinement, and future optimization of both the robot structure and motion planning strategies.